# Design and Evaluation of Multi-User “Entanglement-Based QKD”Using HAP

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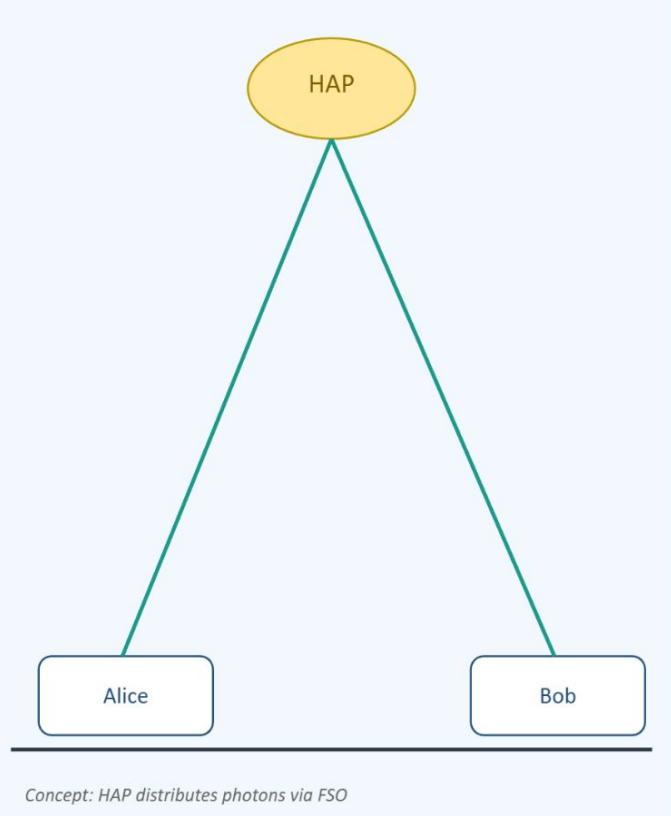
# Introduction

**- QKD** enables secure key sharing using quantum physics

- **HAP (High-Altitude Platform)** acts as an airborne node in the stratosphere

- Uses **free-space optical (FSO)** links instead of fiber

- Goal: **secure key distribution over wide areas**




*Concept: HAP distributes photons via FSO*

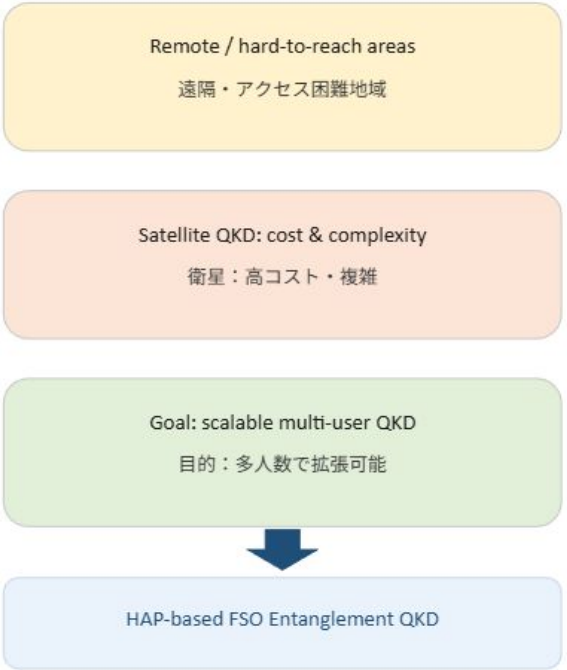
# Motivation

- Long-distance QKD suffers from **high channel loss**
- Multi-user QKD needs **scalability** without many trusted relay nodes
- HAP provides:

  - Better line-of-sight coverage

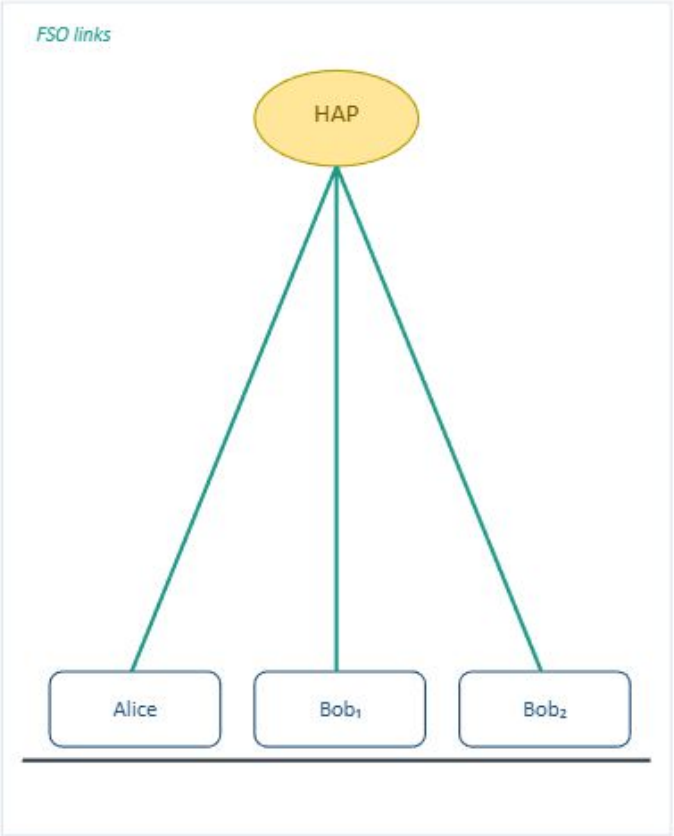
  - Potentially lower cost and easier deployment than satellites

- We want a practical architecture for **multi-user secure key distribution**



# System Model

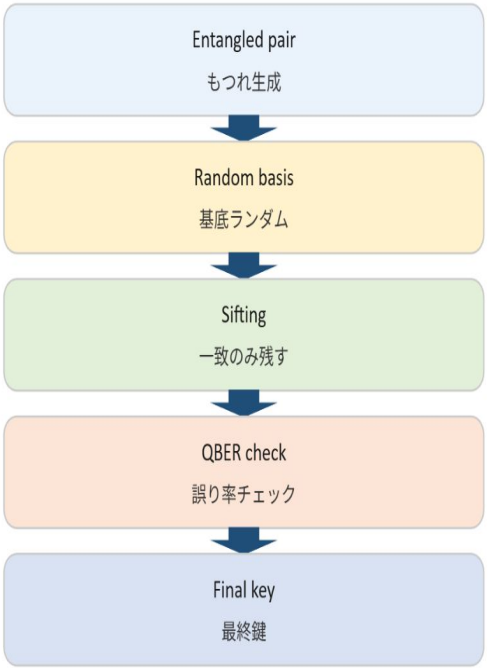
- **HAP** generates **entangled photon pairs (Bell state)**
- One photon → **Alice** (ground server)
- The other photon → **Bob_i** (ground gateway node)
- Communication: **FSO links** (atmosphere + pointing errors affect performance)
- After detection: Alice & Bob compare **basis choices** over a classical channel



# BBM92 Protocol(Entanglement-Based QKD)

- Alice and Bob randomly choose **Z(+)** or **X(×)** basis
- Measure their photons → get raw outcomes
- Publicly share **only the basis**, then **keep matched-basis events** (sifting)
- Estimate **QBER** using a sample; if too high, abort
- Apply **error correction + privacy amplification** → final secret key
- Difference from BB84: **no state preparation by Alice; entanglement source is central**

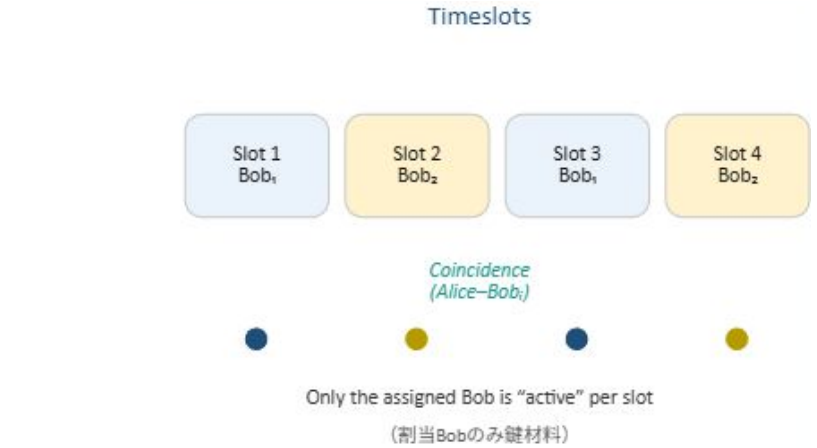
| | 1 | 0 |
|---|---|---|
| + (Z basis) | ↕ | ↔ |
| × (X basis) | ↗ | ↘ |

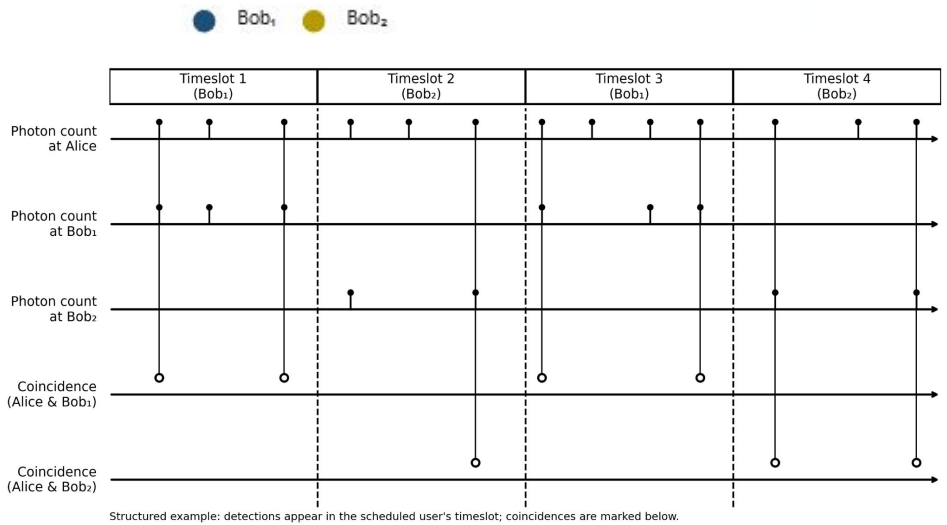



| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Alice's choice of measurement basis | + | × | × | + | + | × | + | × | + | × |
| Alice's measurements | $\|0\rangle$ | $\|+\rangle$ | $\|+\rangle$ | $\|1\rangle$ | $\|0\rangle$ | $\|+\rangle$ | $\|1\rangle$ | $\|-\rangle$ | $\|1\rangle$ | × |
| Bob's choice of measurement basis | × | × | + | × | + | + | × | × | + | + |
| Bob's measurements | $\|+\rangle$ | $\|+\rangle$ | $\|1\rangle$ | $\|-\rangle$ | $\|0\rangle$ | $\|1\rangle$ | $\|+\rangle$ | $\|-\rangle$ | $\|1\rangle$ | $\|0\rangle$ |
| Same basis? (Sifting) | ✗ | ✓ | ✗ | ✗ | ✓ | ✗ | ✗ | ✓ | ✓ | ✗ |
| Shared secret key | | 0 | | | 0 | | | 1 | 1 | |

# TDMA for Multiple Users

- One entangled pair can serve only one Bob at a time
- TDMA splits time into slots and assigns slots to each Bob
- Duaring $Bob_i$'s slot, coincidences are counted only for Alice-$Bob_i$
- Goal: scalable key distribution without adding many intermediate nodes
- Trade-off: more users → less time per user → lower per-user SKR




Structured example: detections appear in the scheduled user's timeslot; coincidences are marked below.

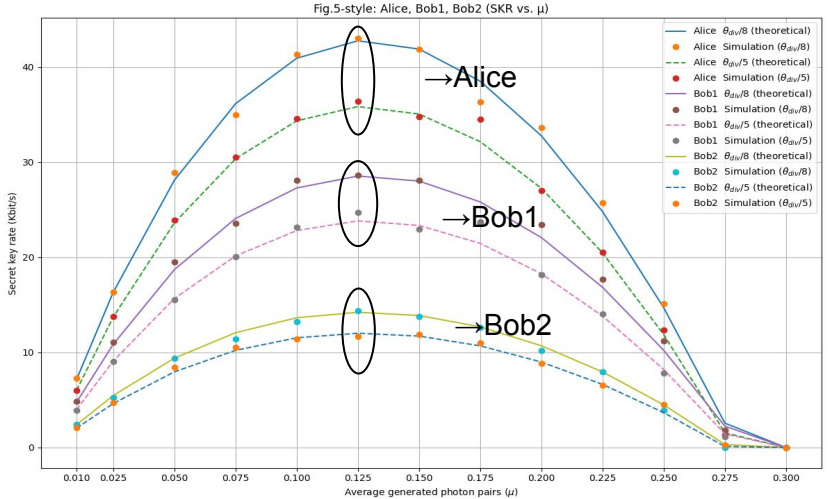
# Results & Validation

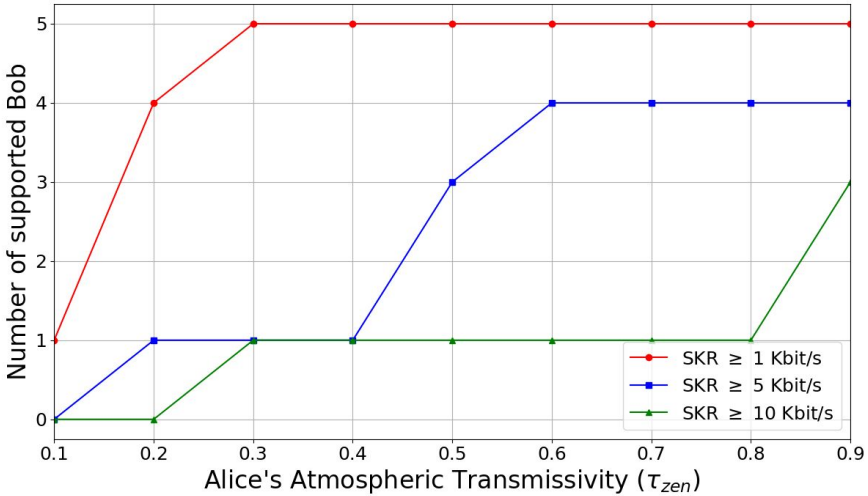
- Evaluated performance using **SKR** and **QBER** under realistic FSO conditions
- Key observation: SKR vs photon-pair rate shows an **optimal peak**

  **- Too small → few coincidences**

  **- Too large → higher errors / multi-pair effects**

- **Multi-user evaluation: supported Bobs depend on channel efficiency and SKR target**



# Next Step: Adaptive TDMA Optimization under Clouds

- **Goal:** Improve total SKR in cloud conditions
- **Approach:** Optimize TDMA based on channel quality (allocate/skip slots)
- **Next:** Build and evaluate with a dynamic cloud model